(1) mean, median, mode for following data

C.I	5-10	10-15	15-20	20-25	25-30	30-35	35-40
frequency	6	8	17	21	15	11	12

Sol:

CI	$x_i = \frac{ul + ll}{2}$	f_i	$f_i x_i$	CF
5-10	7.5	6	45	6
10-15	12.5	8	100	14
15-20	17.5	17	297.5	$\boxed{31}$
$\boxed{20-25}$	22.5	$\boxed{21}$	472.5	52
25-30	27.5	15	412.5	67
30-35	32.5	11	357.5	78
35-40	37.5	12	450	90
		$\sum f_i = 90$	$\sum f_i x_i = 2135$	

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \bar{x}$$

$$\bar{x} = \frac{2135}{90}$$

$$\boxed{\bar{x} = 23.72}$$

Median

$$M = L + \left[\frac{\frac{N}{2} - CF}{f}\right] \times h$$

Let us take median class = 20-25

L = lower limit = 20

h = size of the class = 25-20 = 5

CF = CF preceding median class = 31

$$M = 20 + \left[\frac{(90/2) - 31}{21}\right] \times 5$$

$$= 20 + \left(\frac{\cancel{14}^{2}}{\cancel{21}_{3}}\right) \times 5 = 20 + 3.33$$

$$\boxed{M = 23.33}$$

Mode

$$M_0 = L + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2}\right) \times h$$

L = lower boundary of modal class

f_m = frequency of modal class (with highest frq)

f_1 = frq preceeding modal class

f_2 = freq succeeding modal class

h = size of the modal class.

Modal class = 20-25

$L = 20, \; f_m = 21, \; f_1 = 17, \; f_2 = 15, \; h = 5$

$$Mo = 20 + \left(\frac{21-17}{2(21)-17-15}\right) \times 5$$

$$= 20 + \left(\frac{\cancel{4}^{2}}{\cancel{10}_{\cancel{2}}}\right) \times \cancel{5}$$

$\boxed{Mo = 22}$

(2) Mean & Standard Deviation for the table of 542 members age distribution

Age in years	20-30	30-40	40-50	50-60	60-70	70-80	80-90
members	3	61	132	153	140	51	2

Sol:

Age in years CI	members f	$x_i = \frac{ll+ul}{2}$	$f_i x_i$	x_i^2	$f_i x_i^2$
20-30	3	25	75	625	1875
30-40	61	35	2135	1225	74675
40-50	132	45	5940	2025	267300
50-60	153	55	8415	3025	463425
60-70	140	65	9100	4225	591500
70-80	51	75	3825	5625	286875
80-90	2	85	170	7225	14450
	$\Sigma f_i = 542$		$\Sigma f_i x_i = 29660$		$\Sigma f_i x_i^2 = 1690100$

Mean $\bar{x} = \dfrac{\sum_i f_i x_i}{\sum_i f_i}$

$$\bar{x} = \frac{29660}{542} = 54.723$$

$\boxed{\bar{x} = 54.723}$ years

Standard Deviation

$$\sigma = \sqrt{\frac{\sum_i f_i x_i^2}{\sum_i f_i} - \left(\frac{\sum_i f_i x_i}{\sum_i f_i}\right)^2}$$

$$\sigma = \sqrt{\frac{\sum_i f_i x_i^2}{\sum_i f_i} - (\bar{x})^2}$$

$$\sigma = \sqrt{\frac{1690100}{542} - \left[\frac{29660}{542}\right]^2}$$

$$\sigma = \sqrt{3118.26 - (54.72)^2}$$

$\boxed{\sigma = 11.12}$ years

(3) $P = \{4, 8, 12, 16, 20, 24\}$

$N = 6$

sample size $n = 2$

$$\boxed{\boxed{SD = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}} = \sqrt{\frac{280}{6}} \quad \sigma = 6.83}$$

sampling should be done without replacement

$^{N}C_{n} = {}^{6}C_{2} = 15$

Samples
$$\begin{pmatrix} (4,8) & (4,12) & (4,16) & (4,20) & (4,24) \\ (8,12) & (8,16) & (8,20) & (8,24) & \\ (12,16) & (12,20) & (12,24) & & \\ (16,20) & (16,24) & (20,24) & & \end{pmatrix}$$

sample distribution means

$$\bar{x} = \begin{pmatrix} 6 & 8 & 10 & 12 & 14 \\ 10 & 12 & 14 & 16 & \\ 14 & 16 & 18 & & \\ 18 & 20 & 22 & & \end{pmatrix}$$

mean of SD $\mu_{\bar{x}} = \dfrac{6+8+10+12+14+10+12+14+16+14+16+18+18+20+22}{15}$

$\mu_{\bar{x}} = \dfrac{210}{15} = 14$ $\boxed{\mu_{\bar{x}} = 14}$

$\bar{x} = \dfrac{4+8+12+16+20+24}{6} = 14$

$\boxed{\bar{x} = 14}$

variance of S.D of mean ($\sigma_{\bar{x}}^2$)

$$= \frac{(6-14)^2+(8-14)^2+(10-14)^2+(12-14)^2+(14-14)^2+(10-14)^2+(12-14)^2+(14-14)^2+(16-14)^2+(14-14)^2+(16-14)^2+(18-14)^2+(18-14)^2+(20-14)^2+(22-14)^2}{15}$$

$$= \frac{(8)^2+(6)^2+(4)^2+(2)^2+(0)+(4)^2+(2)^2+(0)+(2)^2+(0)+(2)^2+(4)^2+(4)^2+(6)^2+(8)^2}{15}$$

$$= \frac{64+36+16+4+16+4+4+4+16+16+36+64}{15}$$

$$\sigma_{\bar{x}}^2 = \frac{280}{15} \qquad \boxed{\sigma_{\bar{x}}^2 = 18.66}$$

$$\boxed{\sigma_{\bar{x}} = 4.32}$$

$\therefore$ i) mean of the population = 14

ii) SD of population = 6.82

iii) mean of sampling distribution of means = 14

iv) standard deviation of the sampling distribution

$\sigma_{\bar{x}} = 4.32$

(4) Given

gain in weights (in kgs) of pigs fed on two diets A & B.

Sol

Diet A 25 32 30 34 24 14 32 24 30 31 35 25

Diet B 44 34 22 10 47 31 40 30 32 35 18 21 35 29 22

Condition to prove,

two diets differ significantly regards their effect on increase in weight.

Null Hypothesis H_0:

mean weight gain for Diet A equals to mean weight gain of Diet B

$\boxed{H_0 : \mu_A = \mu_B \quad (T.T.T)}$

Alternative Hypothesis H_a:

$H_a : \mu_A \neq \mu_B$

for Diet A

$n_A = 12$

$\bar{x}_A = \frac{\Sigma_i x_A}{n_A} = \frac{336}{12} = 28$

$\sigma_A^2 = \frac{\Sigma_i (x_A - \bar{x}_A)^2}{n_A - 1} = \frac{380}{11} = 34.55$

$n_B = 15$

$$\bar{x}_B = \frac{\sum x_B}{n_B} = \frac{450}{15} = 30$$

$$\sigma_B^2 = \frac{\sum (x_B - \bar{x}_B)^2}{n_B - 1} = \frac{1410}{14} = 100.71$$

$n < 30$

Let us go with t tables

$$t_{cal} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= \frac{28 - 30}{s\sqrt{\frac{1}{12} + \frac{1}{15}}}$$

$$s^2 = \frac{(n_A - 1) s_A^2 + (n_B - 1) s_B^2}{(n_A + n_B - 2)}$$

$$= \frac{(11)(34.55) + (14)(100.71)}{25}$$

$s^2 = 71.59$

$\boxed{s = 8.46}$

$$t_{cal} = \frac{-2}{8.46 \times 0.387} = -0.61 \quad \boxed{t_{cal} = -0.61}$$

$t_{tab} : t_{(\alpha/2,\ n_A+n_B-2)}$

Let $\alpha/2 = 0.025$ $\quad \alpha = 0.05$

$t_{(\alpha/2,\ n_A+n_B-2)} = t_{(0.025, 25)}$

$t_{(0.025, 25)} = 2.06$

Conclusion

if $|t_{cal}| > t_{table}$ reject H_0

if $|t_{cal}| \leq t_{tab}$ accept H_0

According to the problem

$0.61 < 2.06$ Accept H_0

∴ According to Null hypothesis there is no significant difference between the two diets regarding increase in weight.

---

(5) A die is thrown 100 times. Test whether die is unbiased at 5% level (Goodness of fit).

No on the die	1	2	3	4	5	6
Frequency	10	12	24	16	20	18

Sol: for Goodness of fit test we can go with Chi square Testing (χ^2)

Given to check,

Null Hypothesis H_0: The given die is unbiased

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

Alternative Hypothesis H_1: The given die is biased

The observed frequencies not fit the expected frequencies

Given obv frequencies $(O_i) = 10, 12, 24, 16, 20, 18$

expected frequency $(E_i) = \frac{O_i}{n} = \frac{100}{6} = 16.67$

Face	O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
1	10	16.67	−6.67	44.48	2.67
2	12	16.67	−4.67	21.8	1.31
3	24	16.67	7.33	53.72	3.23
4	16	16.67	−0.67	0.44	0.03
5	20	16.67	3.33	11.08	0.67
6	18	16.67	1.33	1.76	0.11

$$\sum \frac{(O_i - E_i)^2}{E_i} = 8.02$$

$$\chi^2_{cal} = \sum \frac{(O_i - E_i)^2}{E_i} = 8.02$$

$$\boxed{\chi^2_{cal} = 8.02}$$

$\chi^2_{tab} = \chi^2_{(\alpha, n-1)}$ $\boxed{\text{Given } \alpha = 0.05}$

$= \chi^2_{(0.05, 6-1)}$

$= \chi^2_{(0.05, 5)}$

$\boxed{\chi^2_{tab} = 11.07}$

Conclusion,

If $\chi^2_{cal} > \chi^2_{tab} \rightarrow$ Reject H_0 (die is biased)

If $\chi^2_{cal} \leq \chi^2_{tab} \rightarrow$ Accept H_0 die is fair

According to problem

$8.02 < 11.07$ [Accept H_0] die is fair.

$\therefore$ The given die is unbiased at 5% level.

Q Given

no. of coins = 4

no. of trials = 160

observed frequencies

Heads (x)	0	1	2	3	4
Observed (O)	19	54	58	23	6

for Goodness of fit test we can go with Chi square Testing (χ^2)

Null Hypothesis H_0: The coins are unbiased

→ obv freq fit expected binomial distribution for four unbiased coins.

Alternative Hypothesis H_a: The coins are biased.

According to Binomial distribution

$$\boxed{P(X=r) = {}^nC_r\, p^r q^{n-r}}$$

n = no. of coins, r = no. heads. p = prob of getting head $= \frac{1}{2}$

q = prob of not getting head $= \frac{1}{2}$.

$\boxed{N = 160 \text{ no. of trails}}$

Expected frequency $E_k = N \times P(X=r)$

No. of heads	Observed frequency (O_k)	$P(X=r)$	Expected freq $E_k = 160 \times P(X=r)$
0	19	${}^4C_0 (\frac{1}{2})^0 (\frac{1}{2})^4 = 1/16$	$160 \times 1/16 = 10$
1	54	${}^4C_1 (\frac{1}{2})^1 (\frac{1}{2})^3 = \frac{4}{16}$	$160 \times 4/16 = 40$
2	58	${}^4C_2 (\frac{1}{2})^2 (\frac{1}{2})^2 = \frac{6}{16}$	$160 \times \frac{6}{16} = 60$
3	23	${}^4C_3 (\frac{1}{2})^3 (\frac{1}{2}) = \frac{4}{16}$	$160 \times \frac{4}{16} = 40$
4	6	${}^4C_4 (\frac{1}{2})^4 (\frac{1}{2})^0 = \frac{1}{16}$	$160 \times \frac{1}{16} = 10$
	$\sum O_i = 160$		$\sum E_k = 160$

$$\chi^2_{cal} = \sum \frac{(O_r - E_r)^2}{E_r}$$

r	O_r	E_r	$O_r - E_r$	$(O_r-E_r)^2$	$\frac{(O_r-E_r)^2}{E_r}$
0	19	10	9	81	$81/10 = 8.1$
1	54	40	14	196	$196/40 = 4.9$
2	58	60	-2	4	$4/60 = 0.06$
3	23	40	-17	289	$289/40 = 7.22$
4	6	10	-4	16	$16/10 = 1.6$
					$\sum \frac{(O_r-E_r)^2}{E_r} =$

$$\chi^2_{cal} = \sum \frac{(O_r - E_r)^2}{E_r} = 21.89$$

$$\boxed{\chi^2_{cal} = 21.89}$$

$\chi^2_{tab} = \chi^2_{(\alpha, n-1)}$ Let $\alpha = 0.05$

$n = 5$

$= \chi^2_{(0.05, 4)}$

$$\boxed{\chi^2_{tab} = 9.488}$$

Conclusion,

If $\chi^2_{cal} \leq \chi^2_{tab}$ Accept H_0

If $\chi^2_{cal} > \chi^2_{tab}$ Reject H_0

According to the problem

$21.89 > 9.48$

Since, the null hypothesis is been rejected.

$\therefore$ The coins are biased.

---

(7)

Method-1	20	16	26	27	23	22	
Method-2	27	33	42	35	32	34	38

check the variances of time distribution from population from which these samples are drawn not differ significantly.

Sol: The distribution is on variance so let us go with F-test

According to problem,

Null Hypothesis H_0 : Population variance are equal

$$\sigma_1^2 = \sigma_2^2$$ $\boxed{(T.T.T)}$

Alternative Hypothesis H_a : Population variance are not equal

$$\sigma_1^2 \neq \sigma_2^2$$

Method 1

x_1 : 20 16 26 27 23 22

$n_1 = 6$

$$\bar{x}_1 = \frac{20+16+26+27+23+22}{6} = \frac{134}{6}$$

$$\boxed{\bar{x}_1 = 22.33}$$

$$\boxed{S_1^2 = \frac{\sum(x_1-\bar{x}_1)^2}{n_1-1}}$$

$$S_1^2 = \frac{(20-22.3)^2+(16-22.33)^2+(26-22.33)^2+(27-22.33)^2+(23-22.33)^2+(22-22.33)^2}{6-1}$$

$$\boxed{S_1^2 = 16.67}$$

Method 2

X_2 : 27 33 42 35 32 34 38

$n_2 = 7$

$$\bar{x}_2 = \frac{27+33+42+35+32+34+38}{7} = \frac{241}{7} = 34.43$$

$$S_2^2 = \frac{(27-34.43)^2+(33-34.43)^2+(42-34.43)^2+(35-34.43)^2+(32-34.43)^2+(34-34.43)^2+(38-34.43)^2}{7-1}$$

$$\boxed{S_2^2 = 22.62}$$

$$F_{ratio} = \frac{S_2^2}{S_1^2} = 1.36 \quad (S_2^2 > S_1^2)$$

$$\boxed{F_{cal} = 1.36}$$

$F_{tab} = F_{\alpha}((n_1-1),(n_2-1))$

$F_{tab} = F_{\alpha}(5,6)$ $\boxed{\text{Let } \alpha = 0.05}$

$= F_{0.05}(5,6)$

$\boxed{F_{tab} = 4.95}$

Conclusion :

If $F_{cal} \leq F_{tab}$ Accept H_0

If $F_{cal} > F_{tab}$ Reject H_0

According to problem,

$1.36 < 4.95$ Accept H_0

$\therefore$ There is no significant difference between the variances of two methods.

They have similar variability.

---

⑧ $y = ab^x$ $y = ax^b$ fitting the curve

x	1	2	3	4	5
y	4	7	10	15	25

i) fitting exponential curve $y = ab^x$

Apply log on both sides

$\log y = \log a + x \log b$

Let

$Y = \log y \quad A = \log a \quad B = \log b$

$\Rightarrow Y = A + Bx \rightarrow ①$

Let the normal equations for ① be

$$\sum Y = nA + B\sum x$$

$$\sum xY = \sum xA + B\sum x^2$$

x	~~$X = \log x$~~	y	$Y = \log Y$
1		4	0.6021
2		7	0.8451
3		10	1
4		15	1.1761
5		25	1.3979
$\sum x = 15$		$\sum y = 61$	$\sum Y = 5.0212$

$\sum x^2 = 55 \qquad \sum xY = 16.8991$

from normal equations $\boxed{n=5}$

$5.0212 = 5 \cdot A + 15B \rightarrow ①$

$16.8991 = 15A + 55B \rightarrow ②$

multiply ② with 3 & subtract ② & ③.

$$15.0636 = \cancel{15A} + 45B$$

$$(-)\ 16.8991 = \cancel{15A} + 55B$$

$$-1.8355 = -10B$$

$$\boxed{B = 0.1835}$$

sub B in eq^n ②

$$5.0212 = 5A + 15(0.1835)$$

$$5.0212 - 2.7525 = 5A$$

$$\boxed{A = 0.45374}$$

w.k.T

$A = \log a$ $\quad\quad$ $B = \log b$

$a = 10^A$ $\quad\quad$ $b = 10^B$

$a = 10^{0.45374}$ $\quad\quad$ $b = 10^{0.1835}$

$a = 2.8427$ $\quad\quad$ $b = 1.52$

Final curve is

$$\boxed{y = 2.84(1.52)^x}$$

(ii) Fitting the power curve $y = ax^b$

Apply log on b.s

$$\log y = \log a + b \log x$$

Let $Y = \log y$ $\quad \log a = A$ $\quad \log x = X$

$$\Rightarrow Y = A + bX \rightarrow ①$$

$$\begin{cases} \Sigma Y = nA + b\Sigma X \\ \Sigma XY = A\Sigma X + b\Sigma X^2 \end{cases} \rightarrow \text{normal eq}^{ns}$$

x	$X = \log x$	y	$Y = \log y$	X^2	XY
1	0	4	0.6021	0	0
2	0.3010	7	0.8451	0.0906	0.2544
3	0.4771	10	1	0.2276	0.4771
4	0.6021	15	1.1761	0.3625	0.7077
5	0.6990	25	1.3979	0.4886	0.9771
	$\Sigma X = 2.0792$		$\Sigma Y = 5.0212$	$\Sigma X^2 = 1.1693$	$\Sigma XY = 2.4163$

$$5.0212 = 5A + 2.0792b \rightarrow ②$$

$$2.4163 = 2.0792A + 1.1693b \rightarrow ③$$

$$② \times 2.0792 - ③ \times 5$$

$$10.4400 = \cancel{10.396A} + 4.3230b$$
$$12.0815 = \cancel{10.396A} + 5.8465b$$

$$-1.6415 = -1.5235b$$

$$\boxed{b = 1.07}$$

sub b in eq^n ②

$$5.0212 = 5A + (2.0792)(1.07)$$

$$2.796 = 5A$$

$$\boxed{A = 0.5592}$$

$$A = \log a$$

$$a = 10^{A}$$

$$\Rightarrow a = 10^{0.5592}$$

$$\boxed{a = 3.624}$$

final curve is

$$\boxed{y = (3.624)(x)^{1.07}}$$

9. coefficient of correlation (r)

x	15	18	20	24	30	35	40	50
y	85	93	95	105	120	130	150	160

Sol correlation coefficient of (x & y) is

$$r = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{\sum (x-\bar{x})^2}\sqrt{\sum (y-\bar{y})^2}}$$

$$\boxed{\bar{x} = \frac{\sum x_i}{n} = \frac{232}{8} = 29} \quad \boxed{\bar{y} = \frac{\sum y_i}{n} = \frac{938}{8} = 117.25}$$

x	$x-\bar{x}$	$(x-\bar{x})^2$	y	$y-\bar{y}$	$(y-\bar{y})^2$	$(x-\bar{x})(y-\bar{y})$
15	-14	196	85	-32.25	1040	451.5
18	-11	121	93	-24.25	588	266.75
20	-9	81	95	-22.25	495	200.25
24	-5	25	105	-12.25	150	61.25
30	1	1	120	2.75	7.56	2.75
35	6	36	130	12.75	162.56	76.5
40	11	121	150	32.75	1072.56	360.25
50	21	441	160	42.75	1827.56	897.75
		$\sum (x-\bar{x})^2 = 1022$			$\sum (y-\bar{y})^2 = 5343.24$	$\sum (x-\bar{x})(y-\bar{y}) = 2317$

sub the values in r

$$r = \frac{2317}{\sqrt{1022}\sqrt{5343.24}}$$

$$= \frac{2317}{(31.9)(73.09)}$$

$$\boxed{r = 0.993}$$

This shows that as X increases Y increases almost perfectly.

(10). Regression lines

x	65	66	67	67	68	69	70	72
y	67	68	65	68	72	72	69	71

Sol:

regression lines for the given data is

(i) y on x (ii) x on y

y on x => $y - \bar{y} = b_{yx}(x - \bar{x})$

x on y => $x - \bar{x} = b_{xy}(y - \bar{y})$

b_{yx}, b_{xy} are regression coefficients

$$b_{yx} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2}, \quad b_{xy} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(y-\bar{y})^2}$$

$$\bar{x} = \frac{\sum x}{n} = \frac{544}{8} = 68$$

$$\bar{y} = \frac{\sum y}{n} = \frac{552}{8} = 69$$

x	$x-\bar{x}$	y	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$	$(x-\bar{x})^2$	$(y-\bar{y})^2$
65	-3	67	-2	6	9	4
66	-2	68	-1	2	4	1
67	-1	65	-4	4	1	16
67	-1	68	-1	1	1	1
68	0	72	3	0	0	9
69	1	72	3	3	1	9
70	2	69	0	0	4	0
72	4	71	2	8	16	4
				$\sum(x-\bar{x})(y-\bar{y}) = 24$	$\sum(x-\bar{x})^2 = 36$	$\sum(y-\bar{y})^2 = 44$

$$b_{yx} = \frac{24}{36} \qquad b_{xy} = \frac{24}{44}$$

$$\boxed{b_{yx} = 0.66} \qquad \boxed{b_{xy} = 0.5454}$$

regression lines are

$$y - 69 = (0.66)(x - 68) \quad [y \text{ on } x]$$

$$x - 68 = (0.5454)(y - 69) \quad [x \text{ on } y]$$

$$\boxed{y = 0.66x + 24.12} \quad y \text{ on } x$$

$$\boxed{x = (0.5454)y + 30.36} \quad x \text{ on } y$$

Given,

obtain the estimate of x for $y = 70$

from the regression lines sub the y value to obtain the value of x

$$x = (0.5454)(70) + 30.36$$

$$\boxed{x = 68.538} \text{ when } \boxed{Y = 70}$$